



2011

Preliminary Course

FINAL EXAMINATION

Monday, September 12th

Mathematics

General Instructions

- o Reading Time - 5 minutes.
- o Working Time – 3 hours.
- o Write using a black or blue pen.
- o Approved calculators may be used.
- o All necessary working should be shown for every question.
- o Begin each question in a new booklet.

Total marks (120)

- o Attempt Questions 1- 10.
- o All questions are of equal value.

Outcomes to be Assessed:

A student:

- P2** provides reasoning to support conclusions which are appropriate to the context.
- P3** performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities.
- P4** chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques.
- P5** understands the concept of a function and the relationship between a function and its graph.

Question 1 (12 Marks)

Use a Separate Booklet

Marks

- a) Evaluate correct to 1 decimal place $\frac{\sqrt[5]{32}}{6.1+3.2}$ **2**
- b) Factorise completely: $xy - 2yz + 4x - 8z$ **2**
- c) Simplify: $\frac{x}{4} - \frac{3x+1}{6}$ **3**
- d) Write 0.00014 in scientific notation **1**
- e) Find the value of b if $11 + 2\sqrt{18} = 11 + \sqrt{b}$ **1**
- f) Factorise $4m^2 - 9$ **1**
- g) Solve for x : $\frac{1}{x} - 3 = \frac{1}{2x}$ **2**

Question 2 (12 Marks)

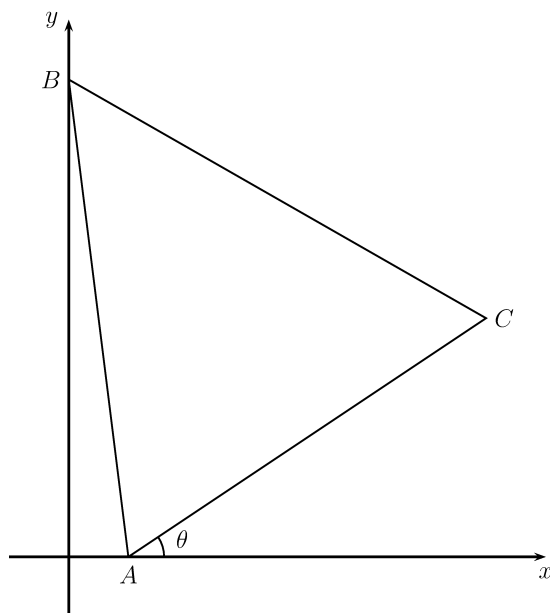
Start a new booklet

Marks

- a) Express $0.4\dot{6}$ as a common fraction. 2
- b) Mr Dandy's water rate for his property has increased by 8%. The new water rate for his property is now \$256. Find the old water rate (correct to the nearest dollar) 2
- c) Solve the inequality $|2 - 5x| < 7$. 2
- d) Express $\frac{1 + \sqrt{3}}{5 - 2\sqrt{3}}$ as a fraction with a rational denominator. 2
- e) Solve the equation $2 \cos \beta = -\sqrt{3}$ for $0^\circ \leq \beta \leq 360^\circ$ 2
- f) Find the exact value of $\tan 120^\circ$. 2

Question 3 (12 marks)

Start a new booklet

Marks

Consider the three points A, B and C on the number plane as shown above.

The equation of the line AC is given by $2x - 3y - 2 = 0$ and the equation of the line BC is given by $4x + 7y - 56 = 0$. The point B has coordinates (0, 8).

- | | | |
|----|---|---|
| a) | Find the coordinates of the point A. | 1 |
| b) | Show that the point C has the coordinates (7, 4). | 2 |
| c) | Find the exact distance of AC in simplest form. | 2 |
| d) | The line AC makes an angle θ with the positive x-axis. Calculate θ to the nearest minute. | 2 |
| e) | Find the equation of the line which is perpendicular to AC and passes through B. | 2 |
| f) | Show that the perpendicular distance from B to AC is $2\sqrt{13}$ | 2 |
| g) | Calculate the area of $\triangle ABC$. | 1 |

Question 4 (12 Marks)

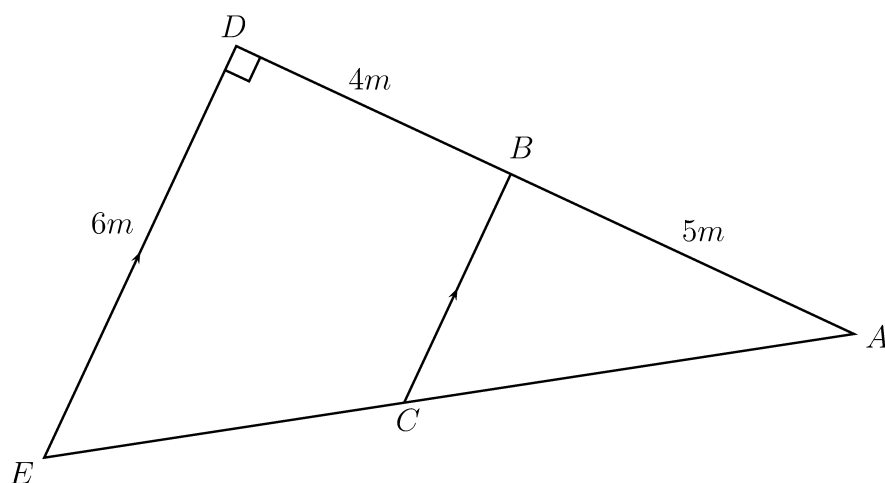
Start a new booklet

Marks

a) Factorise and simplify $\frac{x^3 - 8}{2x^2 - 3x - 2}$

3

- b) In the diagram below triangle ADE is right angled at D . BC is parallel to DE .
 $AB = 5m$, $DB = 4m$ and $DE = 6m$.



- i) Prove that $\triangle ABC$ is similar to $\triangle ADE$.

2

- ii) Find the length of BC

2

- iii) Find the area of $DBCE$.

1

- iv) Show that $AB \times EC = AC \times BD$, giving reasons.

2

- c) State the domain and range of the curve with equation $y = \frac{3}{x} + 1$

2

Question 5 (12 Marks)

Start a new booklet

Marks

- a) The sum of the interior angles of a regular polygon is 2520°
- i) Show that the polygon has sixteen sides. **1**
 - ii) Find the size of each interior angle to the nearest minute. **1**
- b) A given parabola has the equation $y = x^2 - 4x - 12$
- i) By completing the square, write the equation of the parabola in the form $y = (x - h)^2 + k$ **1**
 - ii) State the coordinates of the vertex. **1**
 - iii) Hence or otherwise find the x-intercepts of the parabola $y = x^2 - 4x - 12$. **2**
 - iv) Sketch the parabola showing all relevant features. **2**
- c) The equation $2x^2 - 7x + 12 = 0$ has roots α and β . Find the value of:
- i) $\alpha + \beta$ **1**
 - ii) $\alpha\beta$ **1**
 - iii) $\alpha^2 + \beta^2$ **2**

Question 6 (12 Marks)

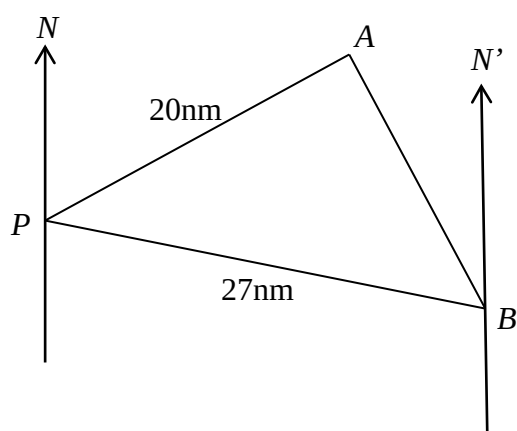
Start a new booklet

Marks

a) Find the value of a if $\tan 15^\circ = \cot(a + 30)^\circ$ **1**

b) Solve for p : $2^{p+2} = 16$ **2**

- c) A ship A has sailed 20 nautical miles from a port P on a bearing of 055° . Ship B has sailed 27 nautical miles from port P on a bearing of 115° .



- i) Copy the diagram above and mark all given information. **1**

- ii) Show that the angle $\angle APB = 60^\circ$. **1**

- iii) Calculate the distance between the two ships to the nearest nautical mile. **2**

- iv) Show that $\angle ABP = 46^\circ$ correct to the nearest degree. **2**

- v) Find the bearing of A from B. **1**

d) If $f(x) = x^2$ and $g(x) = 2x + 1$, find $f(g(2))$. **2**

Question 7 (12 Marks)

Start a new booklet

Marks

- a) Solve $4^x - 9(2^x) + 8 = 0$ **3**
- b) Consider two functions $f(x)$ and $g(x)$.
- i) If $f(x) = g(x) + g(-x)$ show that $f(x)$ is an even function. **1**
- ii) Is $f(x) = g(x) - g(-x)$ also an even function? Give reasons for your answer. **1**
- c) i) Sketch the graph of $y = |x - 4| - 2$ showing all important features. **2**
- ii) On the same number plane shade the region where $y \geq |x - 4| - 2$ and $y < 2$ **2**
- d) Determine the value of a so that $y = ax^2 + 2x + a$ is positive definite. **3**

Question 8 (12 Marks)

Start a new booklet

Marks

- a) If $\tan \theta = \frac{7}{24}$ and $180 \leq \theta \leq 360^\circ$, find the exact value of $\sin \theta$

2

- b) A function is defined by the rule:

$$f(x) = \begin{cases} \frac{1}{x}, & x < 0 \\ x + 2, & x \geq 0 \end{cases}$$

Find:

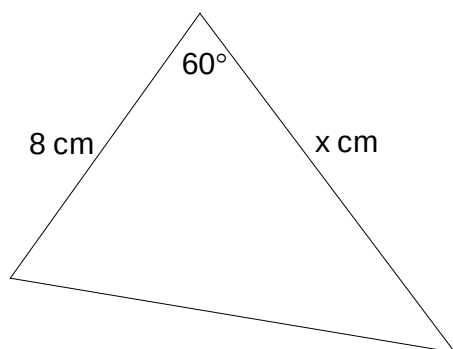
(i) $f(-3)$

1

(ii) $f(b^2)$

1

- c) Calculate the value of x in the diagram below, if the area of the triangle is $22\sqrt{3} \text{ cm}^2$.

2

- d) The equation of a circle is $(x-1)^2 + (y+2)^2 = r^2$

- (i) Write down the coordinates of the centre of this circle.

1

- (ii) If $x + y = 4$ is a tangent to this circle, find the exact value of r .

2

- e) Show that $\tan \theta - \frac{\sin^3 \theta}{\cos \theta} = \sin \theta \cos \theta$

3

Question 9	(12 Marks)	Start a new booklet	Marks
a)	$ABCD$ is a quadrilateral with external angles $\alpha, \beta, \delta, \varphi$. Explain why $\sin(\alpha + \beta + \delta + \varphi) = 0$.		2
b)	i)	Sketch the graph $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$	1
	ii)	On the same number plane sketch $y = \frac{1}{2}$	1
	iii)	How many solutions are there for $\cos x = \frac{1}{2}$ in the domain $0^\circ \leq x \leq 360^\circ$	1
	iv)	For what values of k will $\cos x = k$ have no solutions in the domain $0^\circ \leq x \leq 360^\circ$?	1
c)	Solve $4\sin^2 \theta - 3 = 0$ for θ , where $0^\circ \leq \theta \leq 360^\circ$.		3
d)	Given that $x^2 + 4x + 5 \equiv (x + a)^2 + b^2$ find all possible values for a and b .		3

Question 10 (12 Marks)

Start a new booklet

Marks

a) Find the value(s) of k for which the equation $x^2 + (k+3)x + 4k = 0$ has:

- | | | |
|------|--|---|
| i) | one root equal to 1 | 1 |
| ii) | roots which are equal but opposite in sign | 2 |
| iii) | equal roots | 2 |
| iv) | no real roots | 1 |

b) In the diagram below, $ABCD$ is a square. X , Y and Z are points on the sides AB , BC and CD respectively such that $XB = YC = ZD$.

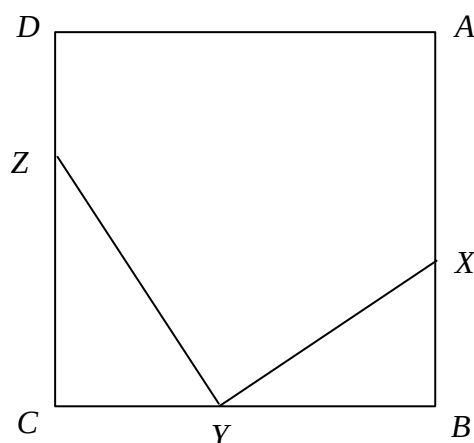


DIAGRAM
NOT TO SCALE

Copy the diagram into your writing booklet.

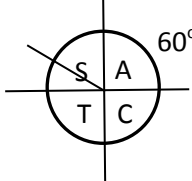
- | | | |
|------|---|---|
| i) | Prove that $\triangle BXY \equiv \triangle CYZ$ | 3 |
| ii) | Hence or otherwise prove that $XY = YZ$ | 1 |
| iii) | Deduce that $\angle XYZ = 90^\circ$ | 2 |

End of Paper

Solutions to Year 11 Mathematics 2011 Preliminary Examination

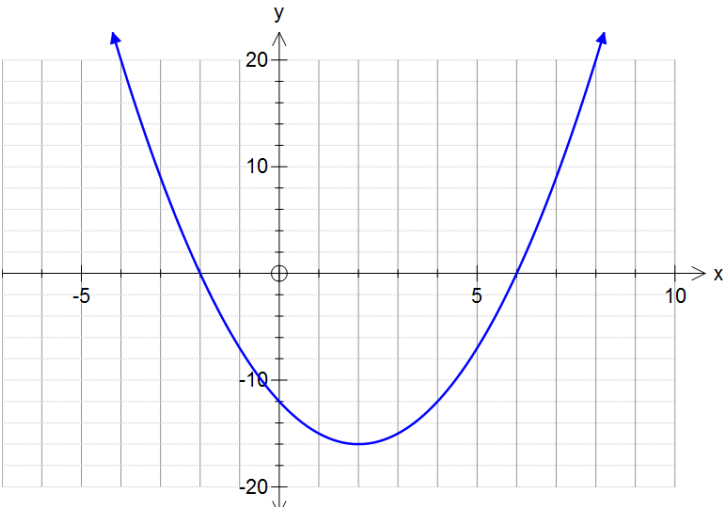
Question 1	Marking Criteria
a) $\frac{\sqrt[5]{32}}{6.1+3.2} = 0.215...$ = 0.2 (1 decimal place)	2 - correct answer 1 - correct calculator read out, but not rounded correctly
$xy - 2yz + 4x - 8z$ b) $= y(x - 2z) + 4(x - 2z)$ $= (y + 4)(x - 2z)$	2 - correct solution 1 - attempt towards solution using some factorising
$\frac{x}{4} - \frac{3x+1}{6}$ c) $= \frac{3x}{12} - \frac{6x+2}{12}$ $= \frac{-3x-2}{12}$ $= -\frac{3x+2}{12}$	3 - correct solution $= \frac{-3x-2}{12}$ 2 - solution with one error 1 - expressing both fractions with a common denominator
d) 1.4×10^{-4}	1 - correct answer
e) $4 \times 18 = 72$ $b = 72$	1 - correct answer
f) $4m^2 - 9$ $= (2m+3)(2m-3)$	1 - correct factorising
$\frac{1}{x} - 3 = \frac{1}{2x}$ g) $2 - 6x = 1$ $-6x = -1$ $x = \frac{1}{6}$	2 - correct solution 1 - correctly removing denominators or one mistake in otherwise correct solution 1 - if multiplying equation through with $2x^2$ and getting the solutions $x = 0$ $x = \frac{1}{6}$ but not realising that the equation is not defined for $x = 0$

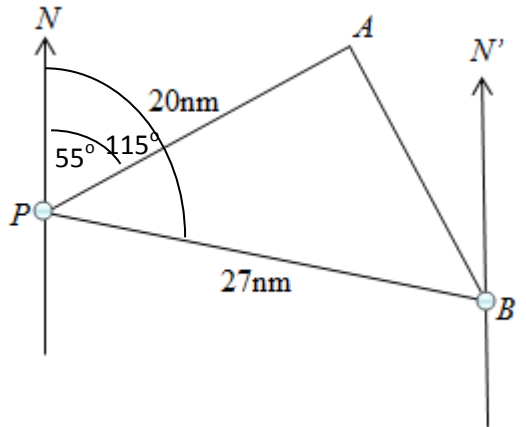
Question 2	
a) let $x = 0.466666\dots$ $10x = 4.666\dots$ $\text{— } x = 0.4666\dots$ $9x = 4.2$ $x = \frac{4.2}{9}$ $x = \frac{42}{90}$ $x = \frac{7}{15}$	2 – correct solution 1 – attempt at correct solution
b) Let $x = \text{old water rate}$ $1.08x = \\$256$ $x = \frac{\\$256}{1.08}$ $= \\$237.07\dots$ $= \\$237(\text{nearest dollar})$	2 – correct solution 1 – attempt at correct solution or answer not rounded to nearest dollar
c) $2 - 5x < 7$ $2 - 5x < 7$ or $-2 + 5x < 7$ $-5x < 5$ $5x < 9$ $x > -1$ $x < \frac{9}{5}$ $\therefore -1 < x < \frac{9}{5}$	2 – correct solution 1 - solution with one error but solution must not be over simplified
d) $\frac{1+\sqrt{3}}{5-2\sqrt{3}} \times \frac{5+2\sqrt{3}}{5+2\sqrt{3}}$ $= \frac{(1+\sqrt{3})(5+2\sqrt{3})}{25-12}$ $= \frac{5+2\sqrt{3}+5\sqrt{3}+6}{13}$ $= \frac{11+7\sqrt{3}}{13}$	2 – correct solution 1 – multiplying with conjugate surd

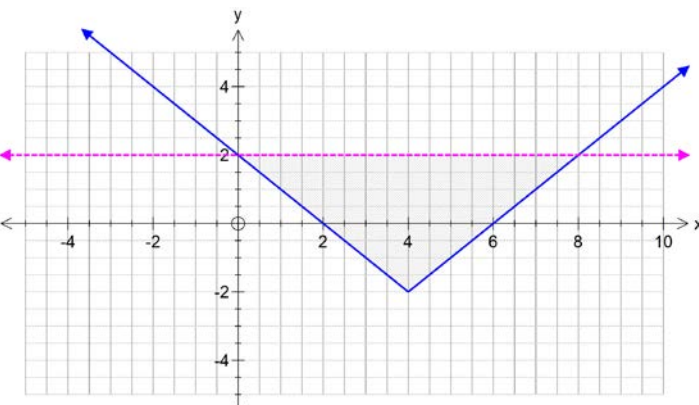
e) $2 \cos \beta = -\sqrt{3}$ $\cos \beta = -\frac{\sqrt{3}}{2}$ $\beta = 180^\circ - 30^\circ, 180^\circ + 30^\circ$ $= 150^\circ, 210^\circ$	2 – correct solution 1 – one angle correct or finding acute angle of 30°
f) $\tan 120^\circ = -\tan 60^\circ$ $= -\sqrt{3}$ 	2 – correct solution 1 – solution with one error
Question3	
a) Let $y = 0$, then $2x - 2 = 0$ $x = 1$ $A = (1, 0)$	1 – correct answer
b) either show C lies on both lines by substitution or solve simultaneously: (1) $2x - 3y - 2 = 0$ (2) $4x + 7y - 56 = 0$ $x = 7, y = 4$	2 – correct solution 1 – solution with one error Or 1 correct substitution
$A = (1, 0) \quad C = (7, 4)$ c) $d = \sqrt{(7-1)^2 + (4-0)^2}$ $d = \sqrt{52}$ $d = 2\sqrt{13}$	2 – correct solution 1 – correct substitution into formula Or correct solution but surd not simplified
$m_{AC} = \frac{2}{3}$ d) $\tan \theta = \frac{2}{3}$ $\theta = 33.69006\dots$ $\theta = 33^\circ 41'$	2 – correct angle to the nearest minute 1 – correct gradient of AC found

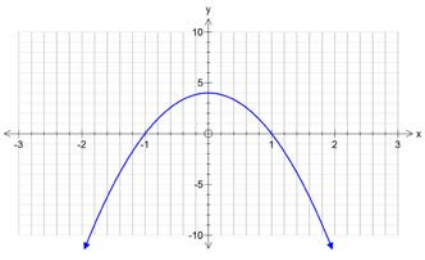
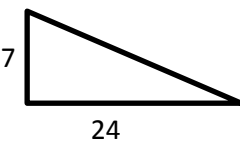
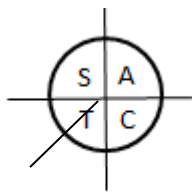
e) $m_{AC} = \frac{2}{3}$, perpendicular gradient $= -\frac{3}{2}$ $B = (0, 8)$, using point gradient form $y - 8 = -\frac{3}{2}(x - 0)$ $2y - 16 = -3x$ $3x + 2y - 16 = 0$	2 – Correct equation of straight line in any form 1 - correct perpendicular gradient
$d = \frac{ 2 \times 0 - 3 \times 8 - 2 }{\sqrt{2^2 + 3^2}}$ $= \frac{ -26 }{\sqrt{13}}$ f) $= \frac{26}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$ $= \frac{26\sqrt{13}}{13}$ $= 2\sqrt{13}$	2 – correct solution 1 – correct substitution into formula
$A = \frac{1}{2} \times 2\sqrt{13} \times 2\sqrt{13}$ $= 26 \text{ units}^2$	1 – correct answer
Question 4	
a) $\frac{x^3 - 8}{2x^2 - 3x - 2}$ $= \frac{(x - 2)(x^2 + 2x + 4)}{(2x + 1)(x - 2)}$ $= \frac{(x^2 + 2x + 4)}{(2x + 1)}$	3 – correctly factorised and simplified 2 – correctly factorised but not simplified 1 – correctly factorised numerator or denominator
b) (i) In triangles $\triangle ABC$ and $\triangle ADE$ $\angle A$ is common $\angle ABC = \angle ADE$ (corresponding angles on parallel lines, $DE \parallel BC$) $\therefore \triangle ABC$ is similar to $\triangle ADE$ (equiangular)	2 – correct proof with all reasons given 1 – correct proof but not all reasons given
ii) $\frac{BC}{6} = \frac{5}{9} \quad (\text{matching sides in similar triangles in the same ratio})$ $BC = \frac{30}{9}$ $= 3.\dot{3}m$	2 – correct solution with correct reason 1 – correct ratio given

iii) $DBCE$ is a trapezium: $A = \frac{4m}{2}(6m + 3.3m)$ $= 18.6m^2$	1 – correct answer
iv) $\frac{AB}{BD} = \frac{AC}{CE}$ (intercepts on transversals cut by parallel lines in same ratio) $\therefore AB \times EC = AC \times BD$	2 – correct sides in ratio statement with correct reason 1 – incorrect reason OR incorrect sides in ratio statement
c) Domain: all real $x, x \neq 0$ Range: all real $y, y \neq 1$	2 – correct domain and range 1 – correct domain or range
Question 5	
a)i) $2520 = (n - 2) \times 180$ $n = 16$	1 – correct solution
ii) $2520 \div 16 = 157^\circ 30'$	1 – correct answer
b)i) $y = x^2 - 4x - 12$ $y = (x^2 - 4x + 4) - 4 - 12$ $y = (x - 2)^2 - 16$	1- correct expression
ii) $V(2, -16)$	1 – correct coordinates
iii) Let $y = 0$ $(x - 2)^2 - 16 = 0$ $(x - 2)^2 = 16$ $(x - 2) = \sqrt{16}$ $x - 2 = \pm 4$ $x = 6$ or $x = -2$	2 – calculating both roots correctly 1 – attempt towards finding roots, or factorising original equation $y = (x - 6)(x - 2)$

iv) 	2 – correct graph showing all intercepts and coordinates of vertex 1 - correct graph with some of the above information
c) i) $\alpha + \beta = \frac{- -7}{2}$ $= \frac{7}{2}$	1 – correct answer
ii) $\alpha\beta = \frac{12}{2}$ $= 6$	1 – correct answer
iii) $\alpha^2 + \beta^2$ $= (\alpha + \beta)^2 - 2\alpha\beta$ $= \left(\frac{7}{2}\right)^2 - 2 \times 6$ $= 0.25$	2 – correct solution 1 – correct algebraic expression using only sum and product, but wrong numbers substituted or mistake made in evaluation
Question 6	
a) Using $\tan 15^\circ = \cot(90 - 15^\circ)$ $a = 45^\circ$	1 – correct answer

b) $2^{p+2} = 16$ $2^{p+2} = 2^4$ $p + 2 = 4$ $p = 2$	2 – correct solution 1 – Writing 16 to the base 2
c) i) 	1 – correct diagram showing sides and bearings
ii) $\angle APB = \angle BPN - \angle APN$ $= 115^\circ - 55^\circ$ $= 60^\circ$	1 – correct answer
iii) Using cosine rule: $x^2 = 20^2 + 27^2 - 2 \times 20 \times 27 \cos 60^\circ$ $x^2 = 589$ $x = 24.26\dots$ $= 24\text{nm (nearest nautical mile)}$	2 – correct solution 1 – correct substitution into formula
iv) Using sine rule: $\frac{\sin \theta}{20} = \frac{\sin 60^\circ}{24}$ $\sin \theta = \frac{\sin 60^\circ}{24} \times 20$ $\sin \theta = 0.72168\dots$ $\theta = \sin^{-1}(0.72168\dots)$ $= 46.194\dots^\circ$ $\theta = 46^\circ \text{ (nearest degree)}$	2 – correct solution 1 – correct substitution into formula

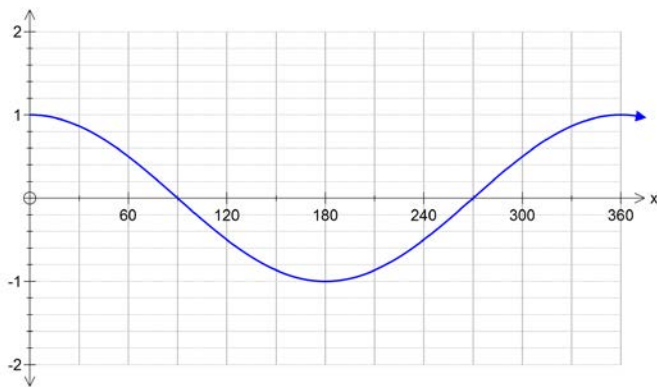
v) $180^\circ - 115^\circ = 65^\circ$ (cointerior angles on parallel lines) $65^\circ - 46^\circ = 19^\circ$ Bearing of A from B: $360^\circ - 19^\circ = 341^\circ$	1 – correct bearing
d) $g(2) = 5$ $f(5) = 25$ $f(g(2)) = 25$	2 – correct answer 1 – finding $g(2)=5$ or applying the function f correctly.
Question 7	
a) $4^x - 9(2^x) + 8 = 0$ $4^x = (2^x)^2$ Let $2^x = u$ $u^2 - 9u + 8 = 0$ $(u - 8)(u - 1) = 0$ $u = 8$ or $u = 1$ $2^x = 8$ $2^x = 1$ $x = 3$ $x = 0$	3 – correct solution 2 – correct substitution and finding of u 1 – correct substitution
b) i) $f(-x) = g(-x) + g(-(-x))$ $= g(-x) + g(x)$ $= g(x) + g(-x)$ $= f(x)$ Hence $f(x)$ is even.	1 – correct answer
$f(-x) = g(-x) - g(-(-x))$ ii) $= g(-x) - g(x)$ $\neq f(x)$ Hence this function is not even (However it is an odd function since $f(-x) = -f(x)$)	1 – correct answer
c)i)ii) 	i) 2 – correct graph showing all intercepts 1 – graph not showing all intercepts or not to scale ii) 2 – line $y=2$ drawn correctly (dotted) and correct shading 1 – line drawn correctly or somewhat correct shading from incorrect line, or line not dotted.

d) positive definite means: $a > 0$ and $\Delta < 0$ $\Delta = 4 - 4a^2$ $4(1 - a^2) < 0$ $(1 - a)(1 + a) < 0$ $a < -1, a > 1$ But $a > 0$ (positive de $\therefore a > 1$ 	3 – correct solution 2 - $\Delta < 0$ solved correctly for a 1 – discriminant found and conditions stated
Question 8	
a) $\tan \theta = \frac{7}{24}$  Using right angled triangle Hypotenuse = $\sqrt{7^2 + 24^2}$ = 25  $\therefore \sin \theta = -\frac{7}{25}$	2 – correct solution 1 – finding the hypotenuse and finding the sine ratio albeit as a positive 1 – correct answer
b)i) $f(-3) = \frac{1}{-3}$ $= -\frac{1}{3}$	1 – correct answer
ii) $f(b^2) = b^2 + 2$ Since $b^2 \geq 0$	1 – correct answer

c) $A = \frac{1}{2}ab \sin C$ $22\sqrt{3} = \frac{1}{2}8x \sin 60^\circ$ $22\sqrt{3} = 4x \frac{\sqrt{3}}{2}$ $22\sqrt{3} = 2x\sqrt{3}$ $x = 11$	2 – correct answer 1 – correct substitution into area formula
d) i) C= (1, -2)	1 – correct coordinates
ii) If $x + y = 4$ is a tangent to this circle, it must touch the circle in one point only. So the radius must be equal to the perpendicular distance from the centre to the line $x + y = 4$ $r = \frac{ 1 \times 1 + 1 \times (-2) - 4 }{\sqrt{1^2 + 1^2}}$ $= \frac{5}{\sqrt{2}}$ $= \frac{5\sqrt{2}}{2}$	2 – correct solution 1 – attempt at solution, substituting into perpendicular distance formula
e) $LHS = \tan \theta - \frac{\sin^3 \theta}{\cos \theta}$ $= \frac{\sin \theta}{\cos \theta} - \frac{\sin^3 \theta}{\cos \theta}$ $= \frac{\sin \theta(1 - \sin^2)}{\cos \theta}$ $= \frac{\sin \theta \cos^2 \theta}{\cos \theta}$ $= \sin \theta \cos \theta$ $= RHS$	3 – correct proof 2 – substituting $\frac{\sin \theta}{\cos \theta} = \tan \theta$ and factorising numerator 1 – substituting $\frac{\sin \theta}{\cos \theta} = \tan \theta$
Question 9	
a) External angles of a polygon add up to 360° $\sin 360^\circ = 0$	2 – correct explanation 1 – realising that external angles add to 360°

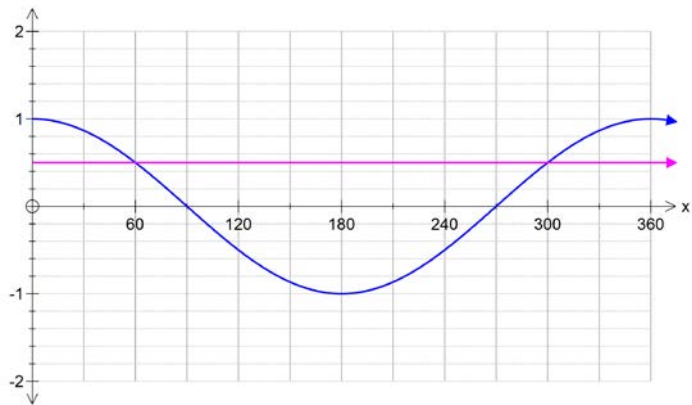
b)

i)



1 – correct graph

ii)



1 – correct line

iii) 2 solutions as there are 2 points of intersection

1 – correct answer

iv) for $k > 1$ or $k < -1$

1 – correct answer

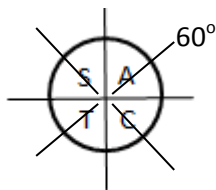
c)

$$4\sin^2 \theta - 3 = 0$$

$$\sin^2 \theta = \frac{3}{4}$$

$$\sin \theta = \pm \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$



3 – correct solution

2 – finding only the solution in the 1st and 2nd quadrant

1 – achieving $\sin^2 \theta = \frac{3}{4}$

d)

$$x^2 + 4x + 5 \equiv (x + a)^2 + b^2$$

$$= x^2 + 2ax + a^2 + b^2$$

$$= x^2 + 2ax + (a^2 + b^2)$$

3 – correct solution

Comparing coefficients $2a = 4 \Rightarrow a = 2$ $a^2 + b^2 = 5 \text{ substituting } a = 2$ $2^2 + b^2 = 5$ $b^2 = 1$ $b = \pm 1$ $\therefore a = 2, b = \pm 1$ Alternatively substitute $x = 0$ and $x = 1$ into identity to give a set of equations that can be solved simultaneously.	2 – solution with one error 1 – realising that coefficient need to be compared and making an attempt to do so.
Question 10	
a)i) substitute $x = 1$ $1 + (k + 3) + 4k = 0$ $5k + 4 = 0$ $k = -\frac{4}{5}$	1 – correct value for k
ii) The sum of roots will equal 0 $\alpha + \beta = 0$ $-(k + 3) = 0$ $-k - 3 = 0$ $k = -3$	2 – correct solution 1 – stating sum of roots equal to 0 OR stating roots as α and $-\alpha$
iii) equal roots means that the discriminant is equal to zero $\Delta = (k + 3)^2 - 4 \times 1 \times 4k$ $(k + 3)^2 - 4 \times 1 \times 4k = 0$ $k^2 + 6k + 9 - 16k = 0$ $k^2 - 10k + 9 = 0$ $(k - 9)(k - 1) = 0$ $\therefore k = 1, k = 9$	2 – correct solution 1 – correct substitution into discriminant

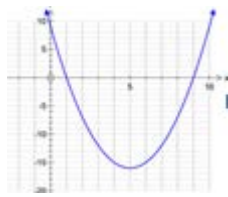
iv) no real roots means that the discriminant is negative

1 – correct solution

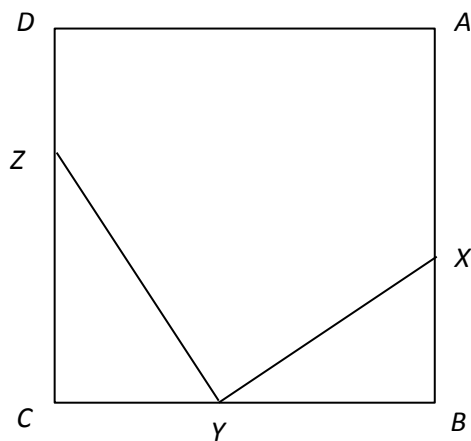
$$\Delta < 0$$

$$(k-9)(k-1) = 0$$

$$1 < k < 9$$



b)



3 – correct proof

2 – correct proof but not all reasons given

1 – one valid statement with a reason

i)

In $\triangle BXY$ and $\triangle CYZ$

$\angle C = \angle B = 90^\circ$ ($ABCD$ is a square)

$XB = YC$ (given)

$ZC = YB$ (since CD and CB are equal sides of a square and $ZD = YC$)

$\therefore \triangle BXY \cong \triangle CYZ$ (SAS)

ii)

$XY = YZ$ (corresponding sides in congruent triangles are equal)

1 – correct reason

iii) Since $\triangle ZCY$ and $\triangle XBY$ are congruent all corresponding angles equal and $\angle ZYC = \angle YXB$ $\angle YXB$ is complementary to $\angle XYB$ (right angled triangle and hence $\angle ZYC$ is complementary to $\angle XYB$ $\angle XYZ = 180^\circ - (\angle ZYC + \angle XYB)$ (straight angle) $\therefore \angle XYZ = 90^\circ$	2 – correct conclusion drawn with all reasons given 1 – correct conclusion but not fully explained.
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Communication Marks:

- Q1 no communication marks
- Q2 1 mark – part b – Set up of solution
Let original = x
 $1.08x = \$256$ etc.
- Q3 1 mark – part b – setting out using LHS /RHS
- Q4 1 mark – part b (iii) – for communicating the area of the shape they are finding or wrote out the formula of the shape they are finding.
- Q5 1 mark – part a (ii) – showing correct formula: angle sum = $(n - 2) \times 180$
1 mark – part b (iv) – stating that “the x-intercepts are -2 and 6”
- Q6 1 mark – part c (ii) – stating that $\angle APB = \angle BPN - \angle APN$
1 mark – part c (iii) – for showing the answer displayed on the calculator before rounding.
1 mark – part c (iv) – for showing “ $\theta = \sin^{-1}$ ” in their solution.
- Q7 1 mark – part b – showing substitution of -x clearly
1 mark – part d – stating condition for positive definite
- Q8 3 marks – part e – clarity of setting out proof with LHS and RHS
- Q9 2 marks – part a – clear statement of exterior angle sum, followed by $\sin 360^\circ = 0$ and hence $\sin(\alpha + \beta + \delta + \gamma) = 0$
- Q10 1 mark – part a – Stating condition for equal/no real roots
1 mark – part b – Giving reasons at each stage of proofs